

Theory of the Affine-DDPNM

Best Approximation, Nested Decomposition, and Error Separation

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This document develops the Galerkin projection theory of the Affine-DDPNM. Starting from the broken-pressure continuous-traction (C-DD) reference system, we prove: (i) the Schur solution at each modal level is the $\hat{\mathbf{S}}^{uu}$ -seminorm best approximation to the full C-DD trace solution; (ii) for nested trace subspaces the error admits an orthogonal Pythagorean decomposition; and (iii) the total volume error is bounded in terms of a C-DD discretisation component and an interface-modal-reduction component. The reduction component is identified exactly in the broken energy and broken- H^1 measures; in L^2 , an exact two-sided triangle bound is combined with a one-sided Poincaré–Friedrichs estimate.

Scope. The reference system throughout is the broken-pressure C-DD formulation (discontinuous P_1 pressure across pores). Equivalence with the monolithic continuous-pressure Taylor–Hood system is not established here; consequently the discretisation term $\|\mathbf{u} - \mathbf{u}_h\|$ appearing in the separation bounds is a C-DD discretisation error, not a standard FEM error. All results assume planar interfaces with a fixed orthonormal frame.

1 C-DD reference system and trace subspaces

1.1 Notation and standing assumptions

We work on a fixed finite-element mesh of the fluid domain $\Omega \subset \mathbb{R}^3$, partitioned into N non-overlapping pore subdomains $\{\Omega_i\}_{i=1}^N$ with m planar internal interfaces $\{\gamma_k\}_{k=1}^m$. On each pore we use the Taylor–Hood $[P_2]^3$ – P_1 pair with **discontinuous** pressure across interfaces (the broken-pressure C-DD formulation). All objects are finite-dimensional; h refers to the given mesh.

Assumption 1.1 (Local algebraic stability). *For every pore i :*

- (a) $A_i \in \mathbb{R}^{n_{u,i} \times n_{u,i}}$ is symmetric positive definite;
- (b) $B_i \in \mathbb{R}^{n_{p,i} \times n_{u,i}}$ has full row rank;
- (c) the constant function 1 belongs to the local pressure space.

For the standard conforming velocity space, Assumption 1.1(a) follows, for example, if Ω_i is connected and Lipschitz and its Dirichlet/no-slip wall portion $\Gamma_{w,i} \subset \partial\Omega_i$ has positive surface measure. These geometric conditions exclude nonzero constant-velocity null modes via the Poincaré–Friedrichs inequality.

Assumption 1.2 (Matching interface trace spaces). *For every internal interface γ_k shared by pores i and j , the two local velocity trace spaces coincide with a shared scalar nodal basis $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ and a one-to-one DOF correspondence between the two sides.*

Dimension conventions. m_k = number of scalar trace DOFs on interface k ; $m_\Gamma = 3 \sum_{k=1}^m m_k$ = total C-DD interface DOFs (three components per trace node). For pore i , $r_i^u = 3 \sum_{\gamma \in \mathcal{U}_i} m_{i,\gamma}$ (C-DD internal trace DOFs) and $r_i^k = 3 \sum_{\gamma \in \mathcal{K}_i} m_{i,\gamma}$ (C-DD boundary trace DOFs).

1.2 Interface frames and affine scalar shapes

On interface γ_k fix an orthonormal frame $\{\mathbf{n}_k, \mathbf{t}_{k,1}, \mathbf{t}_{k,2}\}$ with $\mathbf{n}_k$ directed from the first pore of the pair to the second. Let $\mathbf{c}_k$ be the interface centroid and define the half-span-normalised in-plane coordinates

$$s_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,1}}{a_k}, \quad t_k(\mathbf{x}) = \frac{(\mathbf{x} - \mathbf{c}_k) \cdot \mathbf{t}_{k,2}}{b_k}, \quad (1)$$

where $a_k, b_k > 0$ are the maximal absolute in-plane coordinate values on γ_k . The three affine scalar shape functions are

$$\varphi_0(\mathbf{x}) \equiv 1, \quad \varphi_s(\mathbf{x}) = s_k(\mathbf{x}), \quad \varphi_t(\mathbf{x}) = t_k(\mathbf{x}), \quad (2)$$

and the mode set is $\mathcal{M} := \{0, s, t\} \times \{n, t_1, t_2\}$.

1.3 C-DD coupling, Schur complement, and energy seminorm

For pore i incident on γ_k , define the oriented inward-traction directions $\mathbf{d}_{i,k,\beta}$ ($\beta \in \{n, t_1, t_2\}$) satisfying $\mathbf{d}_{j,k,\beta} = -\mathbf{d}_{i,k,\beta}$ for the opposite pore j .

The C-DD coupling block for direction β on γ_k is

$$[N_{i,k}^\beta]_{\ell,j} = \int_{\gamma_k} \psi_{k,\ell} \mathbf{d}_{i,k,\beta} \cdot \boldsymbol{\phi}_j^{(i)} dS \in \mathbb{R}^{m_k \times n_{u,i}}. \quad (3)$$

Collect the three-direction blocks on all internal ports of pore i into $N_i^u \in \mathbb{R}^{r_i^u \times n_{u,i}}$, and on all boundary ports into $N_i^k \in \mathbb{R}^{r_i^k \times n_{u,i}}$. Let $\boldsymbol{\xi}^k \in \mathbb{R}^{m_\Gamma^k}$ (with $m_\Gamma^k = \sum_i r_i^k$) denote the globally prescribed boundary traction nodal values. Through the boundary affine restriction (see the companion mathematical-properties document, Appendix A) these are determined by the prescribed inlet/outlet pressures.

The velocity response operator is $H_i = A_i^{-1} - A_i^{-1} B_i^\top M_i^{-1} B_i A_i^{-1}$, with $M_i = B_i A_i^{-1} B_i^\top$. H_i is symmetric positive semidefinite and satisfies $H_i^\top A_i H_i = H_i$, $\text{Ker}(H_i) = \text{Range}(B_i^\top)$.

The **internal-unknown block** of the local C-DD Schur complement is

$$\widehat{\mathbf{S}}_i^{uu} = N_i^u H_i (N_i^u)^\top \in \mathbb{R}^{r_i^u \times r_i^u}. \quad (4)$$

Let $\widehat{R}_i^u : \mathbb{R}^{m_\Gamma} \rightarrow \mathbb{R}^{r_i^u}$ be the Boolean restriction extracting the C-DD DOFs of pore i from the global vector of unknown interface DOFs. The assembled C-DD Schur complement (uu-block) is

$$\widehat{\mathbf{S}}^{uu} = \sum_{i=1}^N (\widehat{R}_i^u)^\top \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u \in \mathbb{R}^{m_\Gamma \times m_\Gamma}. \quad (5)$$

$\widehat{\mathbf{S}}^{uu}$ is symmetric positive semidefinite.

The C-DD right-hand side, with prescribed boundary tractions absorbed, is

$$\widehat{\mathbf{F}} = - \sum_{i=1}^N (\widehat{R}_i^u)^\top N_i^u H_i (N_i^k)^\top \widehat{R}_i^k \boldsymbol{\xi}^k \in \mathbb{R}^{m_\Gamma}, \quad (6)$$

where $\widehat{R}_i^k$ extracts the boundary DOFs of pore i from $\boldsymbol{\xi}^k$.

The **full C-DD system** for the unknown interface DOFs is

$$\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}, \quad \boldsymbol{\xi} \in \mathcal{V} := \mathbb{R}^{m_\Gamma}. \quad (7)$$

Assumption 1.3 (Full C-DD consistency). *The assembled full C-DD Schur system (7) is consistent, equivalently $\widehat{\mathbf{F}} \in \text{Range}(\widehat{\mathbf{S}}^{uu})$.*

This assumption is the only global solvability hypothesis used in the projection arguments below. Its derivation from a fully specified broken-pressure mixed Stokes formulation is not repeated in this document. Define the energy seminorm on $\mathcal{V}$ by $\|\boldsymbol{\xi}\|_{\widehat{\mathbf{S}}^{uu}}^2 := \boldsymbol{\xi}^\top \widehat{\mathbf{S}}^{uu} \boldsymbol{\xi}$.

1.4 Restriction operators and nested trace subspaces

Definition 1.4 (Per-interface restriction matrices). For each interface γ_k , let $[M_k]_{\ell,\ell'} = \int_{\gamma_k} \psi_{k,\ell} \psi_{k,\ell'} dS$ be the face mass matrix and $[b_k^\alpha]_\ell = \int_{\gamma_k} \varphi_\alpha \psi_{k,\ell} dS$. Since $\{\psi_{k,\ell}\}_{\ell=1}^{m_k}$ is linearly independent in $L^2(\gamma_k)$, M_k is symmetric positive definite and hence invertible. Define the L^2 -projection vectors $p_k^\alpha = M_k^{-1} b_k^\alpha \in \mathbb{R}^{m_k}$ for $\alpha \in \{0, s, t\}$. For level $\ell \in \{0, 1, 2\}$ with d_ℓ coefficient DOFs per interface,

- $\ell = 0$ (classic P0, $d_0 = 1$): $P_{k,0} = \begin{bmatrix} p_k^0 \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{3m_k \times 1}$;
- $\ell = 1$ (constant-vector P0, $d_1 = 3$): $P_{k,1} = \begin{bmatrix} p_k^0 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & p_k^s & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 3}$;
- $\ell = 2$ (full affine, $d_2 = 9$): $P_{k,2} = \begin{bmatrix} p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & p_k^0 & \mathbf{0} & \mathbf{0} & p_k^s & \mathbf{0} & \mathbf{0} & p_k^t \end{bmatrix} \in \mathbb{R}^{3m_k \times 9}$.

Each column occupies only the block-row matching its direction $\beta \in \{n, t_1, t_2\}$.

The global restriction at level ℓ is the block-diagonal assembly

$$\mathcal{P}_\ell := \bigoplus_{k=1}^m P_{k,\ell} : \mathbb{R}^{d_\ell m} \longrightarrow \mathcal{V}. \quad (8)$$

The corresponding **trace subspace** is $\mathcal{W}_\ell := \text{Range}(\mathcal{P}_\ell) \subset \mathcal{V}$.

Lemma 1.5 (Nested trace subspaces). $\mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \mathcal{W}_2 \subseteq \mathcal{V}$.

Proof. Each $P_{k,0}$ (resp. $P_{k,1}$) is obtained from $P_{k,1}$ (resp. $P_{k,2}$) by selecting a subset of its columns. The block-diagonal structure of $\mathcal{P}_\ell$ preserves the inclusion of ranges. $\square$

Remark 1.6 (Coefficient-space nesting is not literal). *The coefficient spaces $\mathbb{R}^{d_0 m}$, $\mathbb{R}^{d_1 m}$, and $\mathbb{R}^{d_2 m}$ have different dimensions and are not literally nested unless explicit embeddings are specified. The intrinsic nesting used in the Galerkin argument is the nesting of the lifted trace ranges $\mathcal{W}_\ell = \text{Range}(\mathcal{P}_\ell) \subseteq \mathcal{V}$. All results below are stated in terms of these trace subspaces. When we refer to “level ℓ ” we mean the subspace $\mathcal{W}_\ell$ and its associated restriction operator $\mathcal{P}_\ell$.*

2 Best approximation in the energy seminorm

For a fixed level ℓ , the **reduced Schur complement** and **reduced right-hand side** are

$$\mathbf{S}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{S}}^{uu} \mathcal{P}_\ell \in \mathbb{R}^{d_\ell m \times d_\ell m}, \quad \mathbf{F}_\ell = \mathcal{P}_\ell^\top \widehat{\mathbf{F}}. \quad (9)$$

Lemma 2.1 (Reduced consistency). *Under Assumption 1.3, for every level ℓ , $\mathbf{F}_\ell \in \text{Range}(\mathbf{S}_\ell)$.*

Proof. Let $\boldsymbol{\xi}$ be any solution of $\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}$. Set $A_\ell = (\widehat{\mathbf{S}}^{uu})^{1/2} \mathcal{P}_\ell$; then $\mathbf{S}_\ell = A_\ell^\top A_\ell$ and $\mathbf{F}_\ell = A_\ell^\top (\widehat{\mathbf{S}}^{uu})^{1/2} \boldsymbol{\xi}$. Finite-dimensional linear algebra gives $\text{Range}(A_\ell^\top A_\ell) = \text{Range}(A_\ell^\top)$, hence $\mathbf{F}_\ell \in \text{Range}(\mathbf{S}_\ell)$. $\square$

Consequently there exists at least one solution $\boldsymbol{\lambda}_\ell \in \mathbb{R}^{d_\ell m}$ of the reduced system $\mathbf{S}_\ell \boldsymbol{\lambda}_\ell = \mathbf{F}_\ell$. Define the corresponding C-DD trace approximation

$$\boldsymbol{\xi}_\ell := \mathcal{P}_\ell \boldsymbol{\lambda}_\ell \in \mathcal{W}_\ell \subset \mathcal{V}. \quad (10)$$

Let $\boldsymbol{\xi} \in \mathcal{V}$ be any fixed solution of the full C-DD system $\widehat{\mathbf{S}}^{uu} \boldsymbol{\xi} = \widehat{\mathbf{F}}$.

Theorem 2.2 (Galerkin best approximation). *Under Assumptions 1.1, 1.2, and 1.3, for each level $\ell \in \{0, 1, 2\}$,*

$$\|\xi - \xi_\ell\|_{\hat{\mathbf{S}}^{uu}} = \min_{\mu \in \mathbb{R}^{d_{\ell m}}} \|\xi - \mathcal{P}_\ell \mu\|_{\hat{\mathbf{S}}^{uu}}. \quad (11)$$

When $\hat{\mathbf{S}}^{uu}$ is semidefinite the minimiser need not be unique, but the seminorm value is independent of the choice of representatives for both ξ and λ_ℓ .

Proof. From $\hat{\mathbf{S}}^{uu} \xi = \hat{\mathbf{F}}$, $\mathbf{S}_\ell \lambda_\ell = \mathbf{F}_\ell$, $\mathbf{S}_\ell = \mathcal{P}_\ell^\top \hat{\mathbf{S}}^{uu} \mathcal{P}_\ell$, and $\mathbf{F}_\ell = \mathcal{P}_\ell^\top \hat{\mathbf{F}}$, we obtain

$$\mathcal{P}_\ell^\top \hat{\mathbf{S}}^{uu} (\xi_\ell - \xi) = \mathbf{0}. \quad (12)$$

Hence $\mathbf{e}_\ell := \xi - \xi_\ell$ is $\hat{\mathbf{S}}^{uu}$ -orthogonal to $\mathcal{W}_\ell = \text{Range}(\mathcal{P}_\ell)$.

For arbitrary $\mu \in \mathbb{R}^{d_{\ell m}}$, decompose $\xi - \mathcal{P}_\ell \mu = \mathbf{e}_\ell + \mathbf{v}$ with $\mathbf{v} := \xi_\ell - \mathcal{P}_\ell \mu \in \mathcal{W}_\ell$. By (12), $\mathbf{e}_\ell^\top \hat{\mathbf{S}}^{uu} \mathbf{v} = 0$, and the Pythagorean identity in the $\hat{\mathbf{S}}^{uu}$ seminorm yields $\|\xi - \mathcal{P}_\ell \mu\|_{\hat{\mathbf{S}}^{uu}}^2 = \|\mathbf{e}_\ell\|_{\hat{\mathbf{S}}^{uu}}^2 + \|\mathbf{v}\|_{\hat{\mathbf{S}}^{uu}}^2 \geq \|\mathbf{e}_\ell\|_{\hat{\mathbf{S}}^{uu}}^2$. Equality at $\mu = \lambda_\ell$ proves (11).

Semidefinite independence: if $\lambda_\ell, \lambda'_\ell$ both solve $\mathbf{S}_\ell \lambda = \mathbf{F}_\ell$, then $\mathcal{P}_\ell(\lambda_\ell - \lambda'_\ell) \in \text{Ker}(\hat{\mathbf{S}}^{uu})$, so the seminorm on both sides of (11) is unchanged. Similarly, any two full solutions differ by an element of $\text{Ker}(\hat{\mathbf{S}}^{uu})$, and adding an element of this kernel does not change any $\hat{\mathbf{S}}^{uu}$ -seminorm distance. $\square$

3 Nested orthogonal decomposition

Theorem 3.1 (Orthogonal decomposition for nested trace subspaces). *Under the same assumptions as Theorem 2.2, let $\ell < \ell'$, so that $\mathcal{W}_\ell \subseteq \mathcal{W}_{\ell'}$ by Lemma 1.5. Then*

$$\|\xi - \xi_\ell\|_{\hat{\mathbf{S}}^{uu}}^2 = \|\xi - \xi_{\ell'}\|_{\hat{\mathbf{S}}^{uu}}^2 + \|\xi_{\ell'} - \xi_\ell\|_{\hat{\mathbf{S}}^{uu}}^2. \quad (13)$$

Proof. By Theorem 2.2 at level ℓ' , $\mathbf{e}_{\ell'} = \xi - \xi_{\ell'}$ is $\hat{\mathbf{S}}^{uu}$ -orthogonal to $\mathcal{W}_{\ell'}$. The difference $\xi_{\ell'} - \xi_\ell$ satisfies $\xi_{\ell'} \in \mathcal{W}_{\ell'}$ and $\xi_\ell \in \mathcal{W}_\ell \subseteq \mathcal{W}_{\ell'}$, hence $\xi_{\ell'} - \xi_\ell \in \mathcal{W}_{\ell'}$. Therefore $\mathbf{e}_{\ell'} \perp_{\hat{\mathbf{S}}^{uu}} (\xi_{\ell'} - \xi_\ell)$. The decomposition $\xi - \xi_\ell = \mathbf{e}_{\ell'} + (\xi_{\ell'} - \xi_\ell)$ is $\hat{\mathbf{S}}^{uu}$ -orthogonal; the Pythagorean identity gives (13). $\square$

Corollary 3.2 (Three-level gain decomposition). *For $\mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \mathcal{W}_2$,*

$$\|\xi - \xi_0\|_{\hat{\mathbf{S}}^{uu}}^2 = \|\xi - \xi_2\|_{\hat{\mathbf{S}}^{uu}}^2 + \|\xi_2 - \xi_1\|_{\hat{\mathbf{S}}^{uu}}^2 + \|\xi_1 - \xi_0\|_{\hat{\mathbf{S}}^{uu}}^2. \quad (14)$$

Interpretation of the three right-hand-side terms:

- $\|\xi - \xi_2\|_{\hat{\mathbf{S}}^{uu}}^2$: **affine residual** — error after the full nine-mode affine approximation;
- $\|\xi_2 - \xi_1\|_{\hat{\mathbf{S}}^{uu}}^2$: **linear gain** — reduction from adding the six linear modes $\{s, t\} \times \{n, t_1, t_2\}$ to the constant-vector space;
- $\|\xi_1 - \xi_0\|_{\hat{\mathbf{S}}^{uu}}^2$: **tangential gain** — reduction from adding the two tangential constant modes to the single normal constant mode.

Proof. Apply Theorem 3.1 with $(\ell, \ell') = (0, 1)$ and $(\ell, \ell') = (1, 2)$, then substitute. $\square$

Remark 3.3 (Four-space diagnostic diamond). *The three-level chain $\mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \mathcal{W}_2$ combines all six linear modes into one gain term. A controlled diagnostic can instead use four global trace subspaces whose per-interface mode contents are*

$$\begin{aligned} \mathcal{W}_{0n} &: \{1 \cdot n\}, \\ \mathcal{W}_{0v} &: \{1 \cdot n, 1 \cdot t_1, 1 \cdot t_2\}, \\ \mathcal{W}_{1n} &: \{1 \cdot n, s \cdot n, t \cdot n\}, \\ \mathcal{W}_{1v} &: \{1, s, t\} \times \{n, t_1, t_2\}. \end{aligned}$$

Here each notation denotes the range of the corresponding block-diagonal modal lifting assembled over all interfaces, and ξ_{0n} , ξ_{0v} , ξ_{1n} , and ξ_{1v} denote the associated Galerkin trace solutions. The following inclusions always hold: $\mathcal{W}_{0n} \subseteq \mathcal{W}_{0v} \subseteq \mathcal{W}_{1v}$ and $\mathcal{W}_{0n} \subseteq \mathcal{W}_{1n} \subseteq \mathcal{W}_{1v}$. Whether $\mathcal{W}_{0v}$ and $\mathcal{W}_{1n}$ are nonnested depends on the discrete trace space. They are nonnested, for instance, if there exists at least one interface on which the projected linear shapes $\{p_k^s, p_k^t\}$ contain a direction outside $\text{span}\{p_k^0\}$ and the constant tangential liftings are nonzero. The following two nested-path Pythagorean decompositions are valid irrespective of whether either inclusion is strict. Applying Theorem 3.1 gives

$$\begin{aligned}\|\xi - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 &= \|\xi - \xi_{1v}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{1v} - \xi_{0v}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{0v} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2, \\ &= \|\xi - \xi_{1v}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{1v} - \xi_{1n}\|_{\mathbb{S}_{uu}}^2 + \|\xi_{1n} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2.\end{aligned}$$

These are path-dependent marginal-gain decompositions, not a unique causal attribution. Two cleanly defined error-reduction contrasts sharing the same baseline ξ_{0n} are obtained from the Pythagorean identity (Theorem 3.1):

$$\begin{aligned}\Delta_T &:= \|\xi - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 - \|\xi - \xi_{0v}\|_{\mathbb{S}_{uu}}^2 = \|\xi_{0v} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 \geq 0, \\ \Delta_N &:= \|\xi - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 - \|\xi - \xi_{1n}\|_{\mathbb{S}_{uu}}^2 = \|\xi_{1n} - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 \geq 0.\end{aligned}$$

These quantify the reductions in squared Schur-energy error under the two specified nested enrichments. The complementary path-conditional gains are

$$\Delta_{N|T} := \|\xi_{1v} - \xi_{0v}\|_{\mathbb{S}_{uu}}^2, \quad \Delta_{T|N} := \|\xi_{1v} - \xi_{1n}\|_{\mathbb{S}_{uu}}^2.$$

The total reduction from the classic baseline to the full affine solution therefore satisfies

$$\|\xi - \xi_{0n}\|_{\mathbb{S}_{uu}}^2 - \|\xi - \xi_{1v}\|_{\mathbb{S}_{uu}}^2 = \Delta_T + \Delta_{N|T} = \Delta_N + \Delta_{T|N}.$$

In general $\Delta_N \neq \Delta_{N|T}$ and $\Delta_T \neq \Delta_{T|N}$, illustrating that the two paths produce path-dependent marginal-gain decompositions. No identity of the form $\Delta_{\text{total}} = \Delta_T + \Delta_N$ is asserted, and no per-interface additive causal decomposition is claimed.

4 Velocity reconstruction and trace error identity

4.1 Velocity reconstruction from C-DD trace data

For a C-DD trace vector $\zeta \in \mathcal{V}$ (unknown internal DOFs) and prescribed boundary traction data ξ^k , the local velocity field on pore i is

$$U_i(\zeta; \xi^k) = -H_i \left[(N_i^u)^\top \widehat{R}_i^u \zeta + (N_i^k)^\top \widehat{R}_i^k \xi^k \right] \in \mathbb{R}^{n_{u,i}}. \quad (15)$$

The first term couples the unknown interface DOFs to the pore interior; the second term injects the prescribed boundary tractions. The global reconstructed velocity field $\mathbf{u}_h(\zeta)$ is the discontinuous assembly of the local fields $\{U_i(\zeta; \xi^k)\}$ across pores.

Lemma 4.1 (Trace-to-velocity error identity). *For any two trace vectors $\zeta, \zeta' \in \mathcal{V}$ sharing the same prescribed boundary data ξ^k ,*

$$\sum_{i=1}^N \|U_i(\zeta) - U_i(\zeta')\|_{A_i}^2 = \|\zeta - \zeta'\|_{\mathbb{S}_{uu}}^2, \quad (16)$$

where $\|U\|_{A_i}^2 := U^\top A_i U$.

Proof. The boundary coupling terms $(N_i^k)^\top \widehat{R}_i^k \boldsymbol{\xi}^k$ are identical for both arguments and cancel in the difference:

$$U_i(\boldsymbol{\zeta}) - U_i(\boldsymbol{\zeta}') = -H_i(N_i^u)^\top \widehat{R}_i^u(\boldsymbol{\zeta} - \boldsymbol{\zeta}').$$

Using $H_i^\top A_i H_i = H_i$,

$$\begin{aligned} \|U_i(\boldsymbol{\zeta}) - U_i(\boldsymbol{\zeta}')\|_{A_i}^2 &= (\boldsymbol{\zeta} - \boldsymbol{\zeta}')^\top \widehat{R}_i^{u\top} N_i^u H_i A_i H_i (N_i^u)^\top \widehat{R}_i^u (\boldsymbol{\zeta} - \boldsymbol{\zeta}') \\ &= (\boldsymbol{\zeta} - \boldsymbol{\zeta}')^\top \widehat{R}_i^{u\top} \widehat{\mathbf{S}}_i^{uu} \widehat{R}_i^u (\boldsymbol{\zeta} - \boldsymbol{\zeta}'). \end{aligned}$$

Summing over i and using (5) gives (16). $\square$

4.2 Notation for the true solution and its approximations

Let $\mathbf{u}_h := \mathbf{u}_h(\boldsymbol{\xi})$ be the C-DD full-trace reconstruction using a solution $\boldsymbol{\xi}$ of (7). For level ℓ , let $\mathbf{u}_\ell := \mathbf{u}_\ell(\boldsymbol{\xi}_\ell)$. Let $\mathbf{u}$ denote the exact Stokes solution with the prescribed continuous boundary-traction data. The full C-DD reconstruction $\mathbf{u}_h$ and the reduced reconstruction $\mathbf{u}_\ell$ share the same prescribed discrete boundary-traction vector $\boldsymbol{\xi}^k$, obtained from the boundary restriction in (6). Consequently the discrete boundary terms cancel exactly in $\mathbf{u}_h - \mathbf{u}_\ell$. Any discrepancy between the continuous boundary data and its discrete representation is included in the C-DD discretisation error $\mathbf{u} - \mathbf{u}_h$. All velocity error fields considered below satisfy homogeneous wall data.

Let $\Gamma_{w,i}$ denote the no-slip wall portion of $\partial\Omega_i$ and define the broken wall-zero space

$$X_0 := \prod_{i=1}^N \{\mathbf{v}_i \in H^1(\Omega_i)^3 : \mathbf{v}_i|_{\Gamma_{w,i}} = \mathbf{0}\}.$$

All velocity error fields used below belong to X_0 because the exact, full C-DD, and reduced solutions impose the same no-slip wall data. For a (possibly discontinuous) field $\mathbf{v}$ define

$$\|\mathbf{v}\|_A^2 := \sum_{i=1}^N \nu \int_{\Omega_i} \nabla \mathbf{v} : \nabla \mathbf{v} \, dx, \quad (17)$$

$$\|\mathbf{v}\|_{L^2}^2 := \sum_{i=1}^N \int_{\Omega_i} |\mathbf{v}|^2 \, dx, \quad (18)$$

$$|\mathbf{v}|_{H^1}^2 := \sum_{i=1}^N \int_{\Omega_i} \nabla \mathbf{v} : \nabla \mathbf{v} \, dx = \frac{1}{\nu} \|\mathbf{v}\|_A^2. \quad (19)$$

On the full broken product space, the two gradient-based quantities $\|\cdot\|_A$ and $|\cdot|_{H^1}$ are seminorms. If every Ω_i is connected and Lipschitz and $|\Gamma_{w,i}| > 0$, Poincaré–Friedrichs makes $\|\cdot\|_A$ a norm on X_0 . These conditions are assumed whenever the term “A-norm” is used below. For a discrete function $\mathbf{v}_h$ with local coefficient vectors $\{V_i\}$, $\|\mathbf{v}_h\|_A^2 = \sum_i V_i^\top A_i V_i$.

5 Volume error—trace error separation

5.1 Two-sided A-norm separation and the trace-dominated regime

Set

$$D_h := \|\mathbf{u} - \mathbf{u}_h\|_A, \quad E_\ell := \|\mathbf{u}_h - \mathbf{u}_\ell\|_A = \|\boldsymbol{\xi} - \boldsymbol{\xi}_\ell\|_{\widehat{\mathbf{S}}^{uu}}, \quad (20)$$

where the second equality is Lemma 4.1 with $\boldsymbol{\zeta} = \boldsymbol{\xi}$, $\boldsymbol{\zeta}' = \boldsymbol{\xi}_\ell$.

Theorem 5.1 (Two-sided volume-trace bound: A -norm). *Under Assumptions 1.1, 1.2, and 1.3, for each level ℓ ,*

$$|E_\ell - D_h| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_A \leq E_\ell + D_h. \quad (21)$$

Equivalently, $|\|\mathbf{u} - \mathbf{u}_\ell\|_A - E_\ell| \leq D_h$. Hence, whenever $E_\ell > 0$,

$$\left| \frac{\|\mathbf{u} - \mathbf{u}_\ell\|_A}{E_\ell} - 1 \right| \leq \frac{D_h}{E_\ell}. \quad (22)$$

Proof. By the reverse and forward triangle inequalities for the A -norm,

$$|\|\mathbf{u}_h - \mathbf{u}_\ell\|_A - \|\mathbf{u} - \mathbf{u}_h\|_A| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_A \leq \|\mathbf{u}_h - \mathbf{u}_\ell\|_A + \|\mathbf{u} - \mathbf{u}_h\|_A.$$

Substituting E_ℓ and D_h from (20) gives the first statement; dividing by $E_\ell > 0$ yields (22). $\square$

Remark 5.2 (Fixed-mesh trace domination and its limitation). *On a fixed mesh h , define the discretisation-to-reduction ratio*

$$\rho_{\ell,h} := \frac{D_h}{E_{\ell,h}}, \quad (23)$$

*which is well-defined whenever $E_{\ell,h} > 0$. For a prescribed tolerance $0 \leq \eta < 1$, we call the discretisation **trace-dominated at level η** if $\rho_{\ell,h} \leq \eta$. In that case (22) gives the two-sided estimate*

$$(1 - \eta)E_{\ell,h} \leq \|\mathbf{u} - \mathbf{u}_{\ell,h}\|_A \leq (1 + \eta)E_{\ell,h}. \quad (24)$$

This statement is strictly mesh-dependent. It does not imply that mesh refinement leaves the total error unchanged, because both D_h and $E_{\ell,h}$ vary with h . A positive asymptotic lower bound for the total A -error requires control of the discretisation error as well. From (21), $\|\mathbf{u} - \mathbf{u}_{\ell,h}\|_A \geq E_{\ell,h} - D_h$, hence

$$\liminf_{h \rightarrow 0} \|\mathbf{u} - \mathbf{u}_{\ell,h}\|_A \geq \liminf_{h \rightarrow 0} E_{\ell,h} - \limsup_{h \rightarrow 0} D_h.$$

A sufficient condition for a positive asymptotic error floor is therefore

$$\liminf_{h \rightarrow 0} E_{\ell,h} > \limsup_{h \rightarrow 0} D_h. \quad (25)$$

In particular, $D_h \rightarrow 0$ together with $\liminf_{h \rightarrow 0} E_{\ell,h} = c_ > 0$ is sufficient. The present theory does not establish whether (25) holds for the geometries under study; numerical plateaus may motivate such a hypothesis but are not a consequence of these theorems. Moreover, the planar-interface theory developed here does not justify interpreting a plateau on a non-planar watershed partition as evidence for (25).*

Computability. $E_{\ell,h} = \|\boldsymbol{\xi} - \boldsymbol{\xi}_\ell\|_{\mathbb{S}_{uu}}$ is computable when the full C -DD and reduced discrete solutions are both available. In contrast, $D_h = \|\mathbf{u} - \mathbf{u}_h\|_A$ requires the exact PDE solution and is not directly computable in a generic problem; it must be replaced by a manufactured-solution error, an independent overkill reference, or a separate discretisation-error estimator.

5.2 Analogous bounds in L^2 and broken H^1

Theorem 5.3 (Two-sided bound: L^2 -norm). *Under the same assumptions, suppose each pore Ω_i admits a Poincaré–Friedrichs constant $C_{P,i} > 0$ such that $\|\mathbf{v}\|_{L^2(\Omega_i)} \leq C_{P,i} \|\mathbf{v}\|_{H^1(\Omega_i)}$ for all $\mathbf{v}$ vanishing on the wall portion of $\partial\Omega_i$. Set $C_P := \max_i C_{P,i}$ and define*

$$T_\ell^{L^2} := \|\mathbf{u}_h - \mathbf{u}_\ell\|_{L^2}, \quad D_h^{L^2} := \|\mathbf{u} - \mathbf{u}_h\|_{L^2}. \quad (26)$$

Then the following hold.

(a) **Two-sided bound with $T_\ell^{L^2}$:** $|T_\ell^{L^2} - D_h^{L^2}| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_{L^2} \leq T_\ell^{L^2} + D_h^{L^2}$.

(b) **Poincaré upper bound:** $T_\ell^{L^2} \leq \frac{C_P}{\sqrt{\nu}} E_\ell$, and therefore $\|\mathbf{u} - \mathbf{u}_\ell\|_{L^2} \leq D_h^{L^2} + \frac{C_P}{\sqrt{\nu}} E_\ell$.

No corresponding lower bound in terms of E_ℓ is available without a reverse stability estimate (e.g. an inverse inequality bounding the A -norm from above by the L^2 -norm on the discrete velocity space).

Proof. (a) Apply the reverse and forward triangle inequalities for the L^2 -norm to $\mathbf{u} - \mathbf{u}_\ell = (\mathbf{u} - \mathbf{u}_h) + (\mathbf{u}_h - \mathbf{u}_\ell)$. (b) From Lemma 4.1 and per-pore Poincaré–Friedrichs, $(T_\ell^{L^2})^2 = \sum_i \|U_i(\boldsymbol{\xi}) - U_i(\boldsymbol{\xi}_\ell)\|_{L^2(\Omega_i)}^2 \leq \frac{C_P^2}{\nu} \sum_i \|U_i(\boldsymbol{\xi}) - U_i(\boldsymbol{\xi}_\ell)\|_{A_i}^2 = \frac{C_P^2}{\nu} E_\ell^2$. Taking square roots and substituting into the upper bound of (a) gives the second statement. $\square$

Theorem 5.4 (Two-sided bound: broken- H^1 seminorm). *Under the same assumptions, define $T_\ell^{H^1} := |\mathbf{u}_h - \mathbf{u}_\ell|_{H^1}$ and $D_h^{H^1} := |\mathbf{u} - \mathbf{u}_h|_{H^1}$. From (19) and Lemma 4.1, $T_\ell^{H^1} = E_\ell/\sqrt{\nu}$ exactly. Then*

$$|T_\ell^{H^1} - D_h^{H^1}| \leq |\mathbf{u} - \mathbf{u}_\ell|_{H^1} \leq T_\ell^{H^1} + D_h^{H^1} = \frac{1}{\sqrt{\nu}} E_\ell + D_h^{H^1}. \quad (27)$$

Proof. The identity $T_\ell^{H^1} = E_\ell/\sqrt{\nu}$ follows from $|\mathbf{v}|_{H^1}^2 = \|\mathbf{v}\|_A^2/\nu$ and Lemma 4.1. The seminorm triangle inequalities give the two-sided bound. Unlike the L^2 case, both the upper and the lower bound are available here because $T_\ell^{H^1}$ is exactly computable from E_ℓ (no additional inequality needed). $\square$

Remark 5.5 (Pessimism of the L^2 constant). *The constant $C_P = \max_i C_{P,i}$ is a worst-case Poincaré constant and may be pessimistic. For the particular reduction error $\mathbf{e}_\ell := \mathbf{u}_h - \mathbf{u}_\ell$, the field-dependent effective Poincaré ratio is*

$$C_{P,\text{eff}}(\mathbf{e}_\ell) := \frac{\|\mathbf{e}_\ell\|_{L^2}}{|\mathbf{e}_\ell|_{H^1}} = \sqrt{\nu} \frac{\|\mathbf{e}_\ell\|_{L^2}}{\|\mathbf{e}_\ell\|_A},$$

whenever $\mathbf{e}_\ell \neq 0$. Its square is a generalized Rayleigh quotient for the mass and stiffness forms evaluated at this specific error field; it is not a uniform stability constant for the whole space.

6 Discussion

6.1 Summary of the theoretical chain

1. **Best approximation** (Theorem 2.2). Under the explicit consistency assumption for the full C-DD Schur system, the level- ℓ trace solution is a $\hat{\mathbf{S}}^{uu}$ -seminorm best approximation to the full C-DD trace in $\mathcal{W}_\ell$.
2. **Nested decomposition** (Theorem 3.1, Corollary 3.2). For nested modal trace spaces, squared Schur-energy errors satisfy exact Pythagorean identities. The three-level chain therefore separates the classic-P0 reduction error into the residual after full affine approximation and the gains associated with the two nested enrichments.
3. **Volume–trace separation.** In the broken A measure,

$$|E_\ell - D_h| \leq \|\mathbf{u} - \mathbf{u}_\ell\|_A \leq E_\ell + D_h.$$

In the broken- H^1 seminorm the same statement holds after the exact scaling $E_\ell/\sqrt{\nu}$. In L^2 , the exact two-sided triangle bound involves the actual reduction error $T_\ell^{L^2} = \|\mathbf{u}_h - \mathbf{u}_\ell\|_{L^2}$; the Schur-energy error E_ℓ supplies only the upper estimate $T_\ell^{L^2} \leq (C_P/\sqrt{\nu})E_\ell$ unless an additional reverse stability estimate is available.

6.2 Limitations of the current theory

1. **Full C-DD consistency is assumed, not re-proved here.** Assumption 1.3 is sufficient for all Galerkin projection arguments in this document. A self-contained derivation would start from the complete broken-pressure mixed Stokes system and prove that its elimination produces a consistent Schur system.
2. **No C-DD–monolithic equivalence theorem is claimed.** The C-DD reference uses pressure that is discontinuous across pore interfaces, whereas the usual monolithic Taylor–Hood discretisation uses globally continuous P_1 pressure. Equality of their discrete velocity solutions is not established. Even if such an equality were proved, a separate finite-element discretisation error between the monolithic discrete solution and the exact PDE solution would remain.
3. **No mesh-independent positive error floor is proved.** Theorem 5.1 is a fixed-mesh separation result. To infer a refinement-resistant positive floor one would additionally need to control the h -dependence of $E_{\ell,h}$ and D_h ; a sufficient condition is $\liminf_{h \rightarrow 0} E_{\ell,h} > \limsup_{h \rightarrow 0} D_h$ (see (25)). The present theory does not establish whether this condition holds.
4. **The L^2 trace-energy estimate is one-sided.** The Poincaré–Friedrichs inequality bounds $\|\mathbf{u}_h - \mathbf{u}_\ell\|_{L^2}$ from above by E_ℓ , but it yields no lower bound in terms of E_ℓ . Consequently E_ℓ should not be called an L^2 error floor without an additional reverse stability estimate.
5. **The affine residual need not be strictly positive.** Although the nine-mode affine trace space is generally lower dimensional than a nodal C-DD trace space, strict inclusion alone does not imply $E_2 > 0$ for a particular solution. A positive affine residual requires $\xi \notin \mathcal{W}_2 + \text{Ker}(\hat{\mathbf{S}}^{uu})$.
6. **The four-space diamond is diagnostic, not a unique causal decomposition.** When $\mathcal{W}_{0v}$ and $\mathcal{W}_{1n}$ are nonnested, as under the sufficient nondegeneracy condition stated in Remark 3.3, the two Pythagorean paths provide path-indexed marginal-gain decompositions, which are path-dependent in general. Regardless of whether the two middle spaces happen to be comparable for a particular discrete trace space, the quantities Δ_T and Δ_N are controlled error-reduction contrasts associated with specified enrichments, not a unique causal attribution.
7. **Non-planar interfaces are outside the present scope.** Every theorem uses a fixed orthonormal frame on a planar interface. Faceted or genuinely curved interfaces with spatially varying normals introduce a geometric representation issue not covered by these results.

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